

NAME : SHAH FAISAL

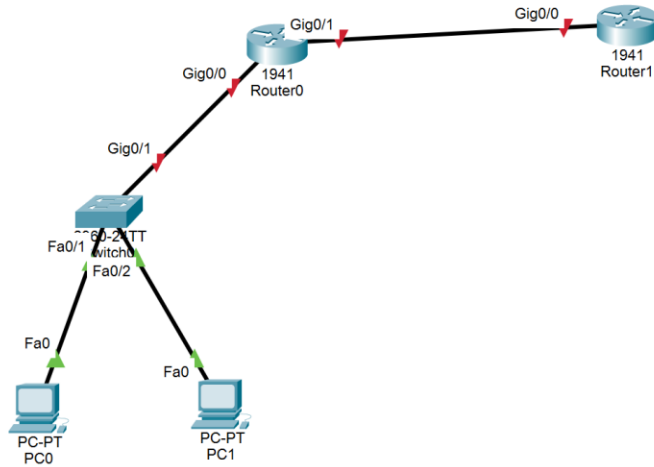
ROLL NO : 23I-0058

SECTION : BSE – 6A

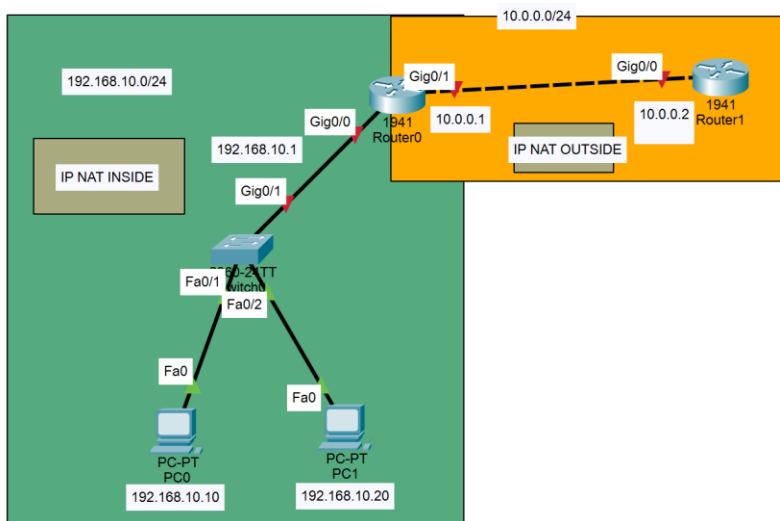
Computer Network
QUIZ 04 AND 05

BSSE SPRING 2026

QUIZ 04 AND 05



Shah Faisal
 23I-0058
 BSE-6A
 QUIZ 04 AND 05



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 QUIZ 04 AND 05

PC0

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 192.168.10.10

Subnet Mask 255.255.255.0

Default Gateway 192.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::202:16FF:FE57:295C

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Root

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PC1

Physical Config **Desktop** Programming Attributes

IP Configuration X

Interface FastEthernet0

IP Configuration

☐ DHCP ☒ Static

IPv4 Address 198.168.10.20

Subnet Mask 255.255.255.0

Default Gateway 198.168.10.1

DNS Server 0.0.0.0

IPv6 Configuration

☐ Automatic ☒ Static

IPv6 Address /

Link Local Address FE80::260:5CFF:FE35:2068

Default Gateway

DNS Server

802.1X

☐ Use 802.1X Security

Authentication MD5

Username

Password

Root

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QUIZ 04 AND 05

 Router0

— □ ×

Physical Config CLI Attributes

IOS Command Line Interface

```
Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to up

Router(config-if)#exit
Router(config)#int gig0/1
Router(config-if)#ip add 10.0.0.1 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/1, changed state to up

Router(config-if)#exit
Router(config)#
Router(config)#
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#we
Translating "we"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#
Router#wr
Building configuration...
[OK]
Router#
```

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☐ Top

IOS Command Line Interface

```
Cisco CISCO1941/K9 (revision 1.0) with 491520K/32768K bytes of memory.
Processor board ID FTX152400KS
2 Gigabit Ethernet interfaces
DRAM configuration is 64 bits wide with parity disabled.
255K bytes of non-volatile configuration memory.
249856K bytes of ATA System CompactFlash 0 (Read/Write)
```

--- System Configuration Dialog ---

Would you like to enter the initial configuration dialog? [yes/no]: no

Press RETURN to get started!

```
Router>en
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0
Router(config-if)#ip add 10.0.0.2 255.255.255.0
Router(config-if)#no shutdown

Router(config-if)#
%LINK-5-CHANGED: Interface GigabitEthernet0/0, changed state to up

%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/0, changed state to
up
Router(config-if)#end
```

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Root

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IOS Command Line Interface

```
Router(config)#
Router(config)#exit
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#we
Translating "we"...domain server (255.255.255.255)
% Unknown command or computer name, or unable to find computer address

Router#
Router#wr
Building configuration...
[OK]
Router#
%LINEPROTO-5-UPDOWN: Line protocol on Interface GigabitEthernet0/1, changed state to
up

Router#
Router#
Router#
Router#
Router#conf t
Enter configuration commands, one per line. End with CNTL/Z.
Router(config)#int gig0/0
Router(config-if)#ip nat inside
Router(config-if)#exit
Router(config)#int gig0/1
Router(config-if)#ip nat outside
Router(config-if)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console
```

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Root

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PC0

Physical Config Desktop Programming Attributes

Command Prompt

```
Reply from 192.168.10.1: bytes=32 time<1ms TTL=255

Ping statistics for 192.168.10.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.1

Pinging 10.0.0.1 with 32 bytes of data:

Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255
Reply from 10.0.0.1: bytes=32 time<1ms TTL=255

Ping statistics for 10.0.0.1:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 0ms, Maximum = 0ms, Average = 0ms

C:\>ping 10.0.0.2

Pinging 10.0.0.2 with 32 bytes of data:

Request timed out.
Request timed out.

Ping statistics for 10.0.0.2:
    Packets: Sent = 2, Received = 0, Lost = 2 (100% loss),

Control-C
^C
C:\>
```

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Router0

Physical Config CLI Attributes

IOS Command Line Interface

```
Enter configuration commands, one per line. End with CNTL/?.
Router(config)#access-list
% Incomplete command.
Router(config)#access-list ?
<1-99>      IP standard access list
<100-199>   IP extended access list
Router(config)#access-list 1 ?
deny        Specify packets to reject
permit      Specify packets to forward
remark      Access list entry comment
Router(config)#access-list 1 permit ?
A.B.C.D     Address to match
any         Any source host
host        A single host address
Router(config)#access-list 1 permit 192.168.10.0
Router(config)#access-list 1 permit 192.168.10.0 ?
A.B.C.D     Wildcard bits
<cr>
Router(config)#access-list 1 permit 192.168.10.0 0.0.0.255
Router(config)#
Router(config)#
Router(config)#
Router(config)#ip nat inside ?
source      Source address translation
Router(config)#ip nat inside source ?
list        Specify access list describing local addresses
static      Specify static local->global mapping
Router(config)#ip nat inside source list ?
<1-199>     Access list number for local addresses
WORD        Access list name for local addresses
Router(config)#ip nat inside source list 1 ?
interface   Specify interface for global address
```

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IOS Command Line Interface

```
Router(config)#ip nat inside ?
  source Source address translation
Router(config)#ip nat inside source ?
  list Specify access list describing local addresses
  static Specify static local->global mapping
Router(config)#ip nat inside source list ?
  <1-199> Access list number for local addresses
  WORD Access list name for local addresses
Router(config)#ip nat inside source list 1 ?
  interface Specify interface for global address
  pool Name pool of global addresses
Router(config)#ip nat inside source list 1 interface ?
  Ethernet IEEE 802.3
  FastEthernet FastEthernet IEEE 802.3
  GigabitEthernet GigabitEthernet IEEE 802.3z
  Serial Serial
Router(config)#ip nat inside source list 1 interface gigabitethernet 0/1 ?
% Unrecognized command
Router(config)#ip nat inside source list 1 interface gigabitethernet 0/1 ?
  overload Overload an address translation
  <cr>
Router(config)#ip nat inside source list 1 interface gigabitethernet 0/1 overload
Router(config)#
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

Router#wr
Building configuration...
[OK]
Router#
```

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Root

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IOS Command Line Interface

```
Serial Serial
Router(config)#ip nat inside source list 1 interface gigabitethernet 0/1 ?
% Unrecognized command
Router(config)#ip nat inside source list 1 interface gigabitethernet 0/1
?
  overload Overload an address translation
  <cr>
Router(config)#ip nat inside source list 1 interface gigabitethernet 0/1
overload
Router(config)#
Router(config)#
Router(config)#end
Router#
%SYS-5-CONFIG_I: Configured from console by console

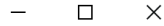
Router#wr
Building configuration...
[OK]
Router#show ip nat translations
Router#
Router#
Router#show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside
global
icmp 10.0.0.1:17          192.168.10.10:17  10.0.0.2:17        10.0.0.2:17
icmp 10.0.0.1:18          192.168.10.10:18  10.0.0.2:18        10.0.0.2:18
icmp 10.0.0.1:19          192.168.10.10:19  10.0.0.2:19        10.0.0.2:19
icmp 10.0.0.1:20          192.168.10.10:20  10.0.0.2:20        10.0.0.2:20

Router#debug ip nat
IP NAT debugging is on
Router#
```

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Router0



Root

Physical Config CLI Attributes

IOS Command Line Interface

```

Router#debug ip nat
IP NAT debugging is on
Router#
NAT: s=192.168.10.10->10.0.0.1, d=10.0.0.2 [21]
NAT*: s=10.0.0.2, d=10.0.0.1->192.168.10.10 [14]
NAT: s=192.168.10.10->10.0.0.1, d=10.0.0.2 [22]
NAT*: s=10.0.0.2, d=10.0.0.1->192.168.10.10 [15]
NAT: s=192.168.10.10->10.0.0.1, d=10.0.0.2 [23]
NAT*: s=10.0.0.2, d=10.0.0.1->192.168.10.10 [16]
NAT: s=192.168.10.10->10.0.0.1, d=10.0.0.2 [24]
NAT*: s=10.0.0.2, d=10.0.0.1->192.168.10.10 [17]
show ip nat translations
Pro  Inside global      Inside local      Outside local      Outside global
icmp 10.0.0.1:21        192.168.10.10:21  10.0.0.2:21       10.0.0.2:21
icmp 10.0.0.1:22        192.168.10.10:22  10.0.0.2:22       10.0.0.2:22
icmp 10.0.0.1:23        192.168.10.10:23  10.0.0.2:23       10.0.0.2:23
icmp 10.0.0.1:24        192.168.10.10:24  10.0.0.2:24       10.0.0.2:24

Router#
NAT: expiring 10.0.0.1 (192.168.10.10) icmp 21 (21)
NAT: expiring 10.0.0.1 (192.168.10.10) icmp 22 (22)
NAT: expiring 10.0.0.1 (192.168.10.10) icmp 23 (23)
  
```

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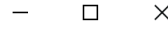
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Realtime

Source | Destination | Type | Color | Time(sec) | Re

Router0



Root  

Physical Config **CLI** Attributes

IOS Command Line Interface

```
Router#show ip nat
% Incomplete command.
Router#show ip nat ?
    statistics      Translation statistics
    translations    Translation entries
Router#show ip nat translations
Router#show ip nat ?
    statistics      Translation statistics
    translations    Translation entries
Router#show ip nat
% Incomplete command.
Router#show ip nat ?
    statistics      Translation statistics
    translations    Translation entries
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 12 Misses: 14
Expired translations: 12
Dynamic mappings:
Router#clear ip nat translations *
^
% Invalid input detected at '^' marker.

Router#clear ip nat translations
^
% Invalid input detected at '^' marker.

Router#clear ip nat translation *
Router#show ip nat translation
Router#
```

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QUIZ 04 AND 05

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Source	Destination	Type	Color	Time(se
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Router0
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Physical
Config
CLI
Attributes

IOS Command Line Interface

```

statistics      Translation statistics
translations    Translation entries
Router#show ip nat translations
Router#show ip nat ?
    statistics      Translation statistics
    translations    Translation entries
Router#show ip nat
% Incomplete command.
Router#show ip nat ?
    statistics      Translation statistics
    translations    Translation entries
Router#show ip nat statistics
Total translations: 0 (0 static, 0 dynamic, 0 extended)
Outside Interfaces: GigabitEthernet0/1
Inside Interfaces: GigabitEthernet0/0
Hits: 12 Misses: 14
Expired translations: 12
Dynamic mappings:
Router#clear ip nat translations *
      ^
% Invalid input detected at '^' marker.

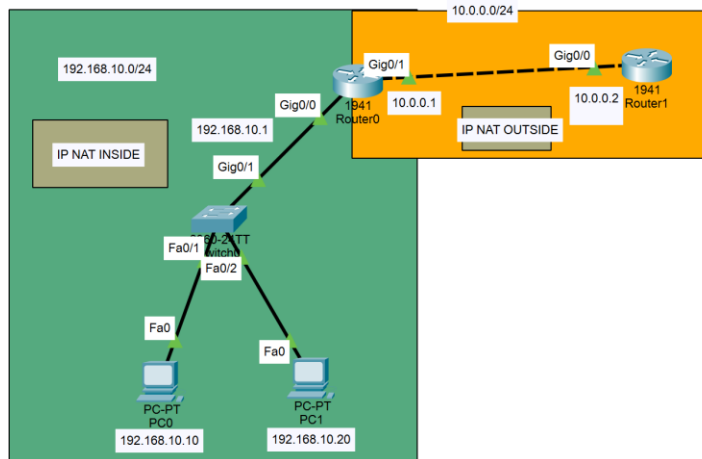
Router#clear ip nat translations
      ^
% Invalid input detected at '^' marker.

Router#clear ip nat translation *
Router#show ip nat translation
Router#wr
Building configuration...
[OK]
Router#
          
```

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Realtime
Simulation

Scenario 0

New
Delete

Toggle PDU List Window

Fire	Last Status	Source	Destination	Type	Color	Time(sec)	Periodic	Num	Edit
Successful		PC0	Router1	IC...		0.000	N	0	(e...
Successful		PC0	Router1	IC...		0.000	N	1	(e...

a) debug ip nat

This command enables real-time logging of NAT translation events, showing each packet's source/destination IP before and after translation. It helps troubleshoot PAT by confirming if traffic is being translated correctly.

b) show ip nat translations

This command displays the active NAT/PAT translation table, showing the mapping between inside local, inside global, outside local, and outside global addresses and ports.

c) Explanation of one output from show ip nat translations

Example output:

text

Pro	Inside global	Inside local	Outside local	Outside global
icmp	203.0.113.1:1024	192.168.1.10:1024	8.8.8.8:1024	8.8.8.8:1024

Detailed explanation:

This entry shows an ICMP (ping) translation where the internal host 192.168.1.10 using source port 1024 is mapped to the public IP 203.0.113.1 with the same port 1024 (PAT overload). The destination is 8.8.8.8. The matching inside local and inside global ports confirm this is a dynamic PAT entry created when the host initiated traffic. Once the ping completes, this entry will eventually time out.

CHATGPT WAS USED FOR HELP

How to configure PAT on Cisco router in Packet Tracer step by step

Configure PAT (Port Address Translation) on a Cisco Router in Packet Tracer — Step by Step

PAT is also called **NAT Overload**. It allows multiple private IP devices to share **one public IP address**.

Example Topology

- PC1 = 192.168.1.2
- PC2 = 192.168.1.3
- Router LAN Interface = 192.168.1.1
- Router WAN Interface = 200.1.1.1

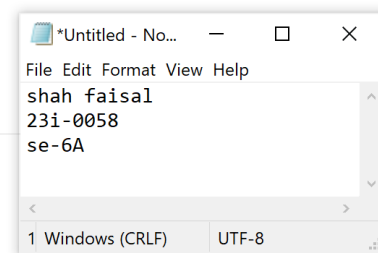
Step 1: Build Topology in Packet Tracer

Use:

- 1 Router
- 1 Switch
- 2 PCs

Connect:

- PCs → Switch
- Switch → Router G0/0
- Router G0/1 → Cloud/ISP



Why is show ip nat translations not showing any output

`show ip nat translations` showing **no output** usually means **PAT/NAT has not been triggered yet** or the configuration is incomplete. NAT entries are created **only when traffic passes through the router from inside to outside**.

Most Common Reasons (in order)

1. No traffic has gone through NAT yet

After configuring PAT, you must generate traffic.

Do this:

From a PC inside network:

```
</> Bash
ping 8.8.8.8
```

or ping the outside device/server.

Then check again:

```
</> Bash
show ip nat translations
```

